Bayesian Tracking: The three main steps

- **Prediction:** we have seen $\boldsymbol{y}_0, \ldots, \boldsymbol{y}_{i-1}$ — what state does this set of measurements predict for the i'th frame? to solve this problem, we need to obtain a representation of $P(\boldsymbol{X}_i | \boldsymbol{Y}_0 = \boldsymbol{y}_0, \ldots, \boldsymbol{Y}_{i-1} = \boldsymbol{y}_{i-1})$.
- **Data association:** Some of the measurements obtained from the i-th frame may tell us about the object's state. Typically, we use $P(\boldsymbol{X}_i | \boldsymbol{Y}_0 = \boldsymbol{y}_0, \ldots, \boldsymbol{Y}_{i-1} = \boldsymbol{y}_{i-1})$ to identify these measurements.
- **Correction:** now that we have $\boldsymbol{y}_i$ — the relevant measurements — we need to compute a representation of $P(\boldsymbol{X}_i | \boldsymbol{Y}_0 = \boldsymbol{y}_0, \ldots, \boldsymbol{Y}_i = \boldsymbol{y}_i)$.

Simplifying Assumptions

- **Only the immediate past matters:** formally, we require

$$P(\boldsymbol{X}_i|\boldsymbol{X}_1,\ldots,\boldsymbol{X}_{i-1}) = P(\boldsymbol{X}_i|\boldsymbol{X}_{i-1})$$

  This assumption hugely simplifies the design of algorithms, as we shall see; furthermore, it isn't terribly restrictive if we're clever about interpreting $\boldsymbol{X}_i$ as we shall show in the next section.

- **Measurements depend only on the current state:** we assume that $\boldsymbol{Y}_i$ is conditionally independent of all other measurements given $\boldsymbol{X}_i$. This means that

$$P(\boldsymbol{Y}_i,\boldsymbol{Y}_j,\ldots\boldsymbol{Y}_k|\boldsymbol{X}_i) = P(\boldsymbol{Y}_i|\boldsymbol{X}_i)P(\boldsymbol{Y}_j,\ldots,\boldsymbol{Y}_k|\boldsymbol{X}_i)$$

  Again, this isn't a particularly restrictive or controversial assumption, but it yields important simplifications.

Base case

Firstly, we assume that we have $P(\boldsymbol{X}_0)$

$$
\begin{aligned}
P(\boldsymbol{X}_0|\boldsymbol{Y}_0 = \boldsymbol{y}_0) &= \frac{P(\boldsymbol{y}_0|\boldsymbol{X}_0)P(\boldsymbol{X}_0)}{P(\boldsymbol{y}_0)} \\
&= \frac{P(\boldsymbol{y}_0|\boldsymbol{X}_0)P(\boldsymbol{X}_0)}{\int P(\boldsymbol{y}_0|\boldsymbol{X}_0)P(\boldsymbol{X}_0)d\boldsymbol{X}_0} \\
&\propto P(\boldsymbol{y}_0|\boldsymbol{X}_0)P(\boldsymbol{X}_0)
\end{aligned}
$$

Induction step

Given $P(\boldsymbol{X}_{i-1}|\boldsymbol{y}_0, \ldots, \boldsymbol{y}_{i-1})$.

Prediction

Prediction involves representing

$$P(\boldsymbol{X}_i|\boldsymbol{y}_0, \ldots, \boldsymbol{y}_{i-1})$$

Our independence assumptions make it possible to write

$$\begin{aligned} P(\boldsymbol{X}_i|\boldsymbol{y}_0, \ldots, \boldsymbol{y}_{i-1}) &= \int P(\boldsymbol{X}_i, \boldsymbol{X}_{i-1}|\boldsymbol{y}_0, \ldots, \boldsymbol{y}_{i-1}) d\boldsymbol{X}_{i-1} \\ &= \int P(\boldsymbol{X}_i|\boldsymbol{X}_{i-1}, \boldsymbol{y}_0, \ldots, \boldsymbol{y}_{i-1}) P(\boldsymbol{X}_{i-1}|\boldsymbol{y}_0, \ldots, \boldsymbol{y}_{i-1}) d\boldsymbol{X}_{i-1} \\ &= \int P(\boldsymbol{X}_i|\boldsymbol{X}_{i-1}) P(\boldsymbol{X}_{i-1}|\boldsymbol{y}_0, \ldots, \boldsymbol{y}_{i-1}) d\boldsymbol{X}_{i-1} \end{aligned}$$

Induction step

Correction

Correction involves obtaining a representation of

$$P(\boldsymbol{X}_i|\boldsymbol{y}_0,\ldots,\boldsymbol{y}_i)$$

Our independence assumptions make it possible to write

$$\begin{aligned}
P(\boldsymbol{X}_i|\boldsymbol{y}_0,\ldots,\boldsymbol{y}_i) &= \frac{P(\boldsymbol{X}_i,\boldsymbol{y}_0,\ldots,\boldsymbol{y}_i)}{P(\boldsymbol{y}_0,\ldots,\boldsymbol{y}_i)} \\
&= \frac{P(\boldsymbol{y}_i|\boldsymbol{X}_i,\boldsymbol{y}_0,\ldots,\boldsymbol{y}_{i-1})P(\boldsymbol{X}_i|\boldsymbol{y}_0,\ldots,\boldsymbol{y}_{i-1})P(\boldsymbol{y}_0,\ldots,\boldsymbol{y}_{i-1})}{P(\boldsymbol{y}_0,\ldots,\boldsymbol{y}_i)} \\
&= P(\boldsymbol{y}_i|\boldsymbol{X}_i)P(\boldsymbol{X}_i|\boldsymbol{y}_0,\ldots,\boldsymbol{y}_{i-1})\frac{P(\boldsymbol{y}_0,\ldots,\boldsymbol{y}_{i-1})}{P(\boldsymbol{y}_0,\ldots,\boldsymbol{y}_i)} \\
&= \frac{P(\boldsymbol{y}_i|\boldsymbol{X}_i)P(\boldsymbol{X}_i|\boldsymbol{y}_0,\ldots,\boldsymbol{y}_{i-1})}{\int P(\boldsymbol{y}_i|\boldsymbol{X}_i)P(\boldsymbol{X}_i|\boldsymbol{y}_0,\ldots,\boldsymbol{y}_{i-1})d\boldsymbol{X}_i}
\end{aligned}$$

Linear dynamic models

- Use notation ~ to mean "has the pdf of", N(a, b) is a normal distribution with mean a and covariance b.
- Then a linear dynamic model has the form

$$\mathbf{x}_i = N\left(\mathbf{D}_{i-1}\mathbf{x}_{i-1}; \Sigma_{d_i}\right)$$

$$\mathbf{y}_i = N\left(\mathbf{M}_i\mathbf{x}_i; \Sigma_{m_i}\right)$$

- This is much, much more general than it looks, and extremely powerful

Examples

- Drifting points
  - we assume that the new position of the point is the old one, plus noise.
  - For the measurement model, we may not need to observe the whole state of the object
    - e.g. a point moving in 3D, at the 3k'th tick we see x, 3k+1'th tick we see y, 3k+2'th tick we see z
    - in this case, we can still make decent estimates of **all three** coordinates at each tick.
  - This property, which does not apply to every model, is called **Observability**

Examples

- Points moving with constant velocity
- Periodic motion
- Etc.
- Points moving with constant acceleration

Points moving with constant velocity

- We have

$$u_i = u_{i-1} + \Delta t v_{i-1} + \varepsilon_i$$
$$v_i = v_{i-1} + \varsigma_i$$

  – (the Greek letters denote noise terms)
- Stack (u, v) into a single state vector

$$\begin{pmatrix} u \\ v \end{pmatrix}_i = \begin{pmatrix} 1 & \Delta t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}_{i-1} + \text{noise}$$

  – which is the form we had above

Points moving with constant acceleration

- We have

$$u_i = u_{i-1} + \Delta t v_{i-1} + \varepsilon_i$$
$$v_i = v_{i-1} + \Delta t a_{i-1} + \varsigma_i$$
$$a_i = a_{i-1} + \xi_i$$

  – (the Greek letters denote noise terms)

- Stack (u, v) into a single state vector

$$\begin{pmatrix} u \\ v \\ a \end{pmatrix}_i = \begin{pmatrix} 1 & \Delta t & 0 \\ 0 & 1 & \Delta t \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} u \\ v \\ a \end{pmatrix}_{i-1} + \text{noise}$$

  – which is the form we had above

The Kalman Filter

- Key ideas:
  - Linear models interact uniquely well with Gaussian noise - make the prior Gaussian, everything else Gaussian and the calculations are easy
  - Gaussians are really easy to represent --- once you know the mean and covariance, you're done

The Kalman Filter in 1D

- Dynamic Model

$$x_i \sim N(d_i x_{i-1}, \sigma_{d_i}^2)$$

- Notation

$$y_i \sim N(m_i x_i, \sigma_{m_i}^2)$$

mean of $P(X_i|y_0, \ldots, y_{i-1})$ as $\overline{X}_i^-$ ——— Predicted mean

Corrected mean — mean of $P(X_i|y_0, \ldots, y_i)$ as $\overline{X}_i^+$

the standard deviation of $P(X_i|y_0, \ldots, y_{i-1})$ as σ_i^-

of $P(X_i|y_0, \ldots, y_i)$ as σ_i^+

Dynamic Model:

$$x_i \sim N(d_i x_{i-1}, \sigma_{d_i})$$

$$y_i \sim N(m_i x_i, \sigma_{m_i})$$

Start Assumptions: $\overline{x_0^-}$ and σ_0^- are known
Update Equations: Prediction

$$\boxed{\begin{aligned} \overline{x_i^-} &= d_i \overline{x}_{i-1}^+ \\ \sigma_i^- &= \sqrt{\sigma_{d_i}^2 + (d_i \sigma_{i-1}^+)^2} \end{aligned}}$$

Update Equations: Correction

$$x_i^+ = \left(\frac{\overline{x_i^-}\,\sigma_{m_i}^2 + m_i y_i (\sigma_i^-)^2}{\sigma_{m_i}^2 + m_i^2 (\sigma_i^-)^2} \right)$$

$$\sigma_i^+ = \sqrt{\left(\frac{\sigma_{m_i}^2 (\sigma_i^-)^2}{(\sigma_{m_i}^2 + m_i^2 (\sigma_i^-)^2)} \right)}$$

Correction for 1D Kalman filter

- Let us look at the equations carefully:

- Notice:
  - if measurement noise is small, we rely mainly on the measurement, if it's large, mainly on the prediction

$$x_i^+ = \left(\frac{\overline{x}_i^- \sigma_{m_i}^2 + m_i y_i (\sigma_i^-)^2}{\sigma_{m_i}^2 + m_i^2 (\sigma_i^-)^2} \right)$$

$$\sigma_i^+ = \sqrt{\left(\frac{\sigma_{m_i}^2 (\sigma_i^-)^2}{(\sigma_{m_i}^2 + m_i^2 (\sigma_i^-)^2)} \right)}$$

In higher dimensions, derivation follows the same lines, but isn't as easy. Expressions here.

Dynamic Model:

$$\boldsymbol{x}_i \sim N(\mathcal{D}_i \boldsymbol{x}_{i-1}, \Sigma_{d_i})$$

$$\boldsymbol{y}_i \sim N(\mathcal{M}_i \boldsymbol{x}_i, \Sigma_{m_i})$$

Start Assumptions: $\overline{\boldsymbol{x}_0^-}$ and Σ_0^- are known

Update Equations: Prediction

$$\overline{\boldsymbol{x}_i^-} = \mathcal{D}_i \overline{\boldsymbol{x}_{i-1}^+}$$

$$\Sigma_i^- = \Sigma_{d_i} + \mathcal{D}_i \sigma_{i-1}^+ \mathcal{D}_i$$

Update Equations: Correction

$$\mathcal{K}_i = \Sigma_i^- \mathcal{M}_i^T \left[\mathcal{M}_i \Sigma_i^- \mathcal{M}_i^T + \Sigma_{m_i}\right]^{-1}$$

$$\overline{\boldsymbol{x}_i^+} = \overline{\boldsymbol{x}_i^-} + \mathcal{K}_i \left[\boldsymbol{y}_i - \mathcal{M}_i \overline{\boldsymbol{x}_i^-}\right]$$

$$\Sigma_i^+ = \left[Id - \mathcal{K}_i \mathcal{M}_i\right] \Sigma_i^-$$

Data Association

- Nearest neighbours
  - choose the measurement with highest probability given predicted state
  - popular, but can lead to catastrophe
- Probabilistic Data Association
  - combine measurements, weighting by probability given predicted state
  - gate using predicted state

Particle Filters

- Have N samples from time $t-1$,
  $S_{t-1} = \{\mathbf{s}_{t-1}^{(i)}, \pi_{t-1}^{(i)}, i = 1, ..., N\}$
- Want to construct S_t
- For $n = 1 : N$,
  1. Select a sample $\mathbf{s}_t^{\prime(n)}$ from S_{t-1} according to probability π_{t-1}
  2. Predict $\mathbf{s}_t^{(n)}$ by sampling from $p(\mathbf{x}_t | \mathbf{x}_{t-1} = \mathbf{s}_t^{\prime(n)})$
  3. Measure and weight the new position according to image features $\mathbf{z}_t^{\mathrm{y}}$,
     $\pi_t^{(n)} = p(\mathbf{z}_t^{\mathrm{y}} | \mathbf{x}_t = \mathbf{s}_t^{(n)})$

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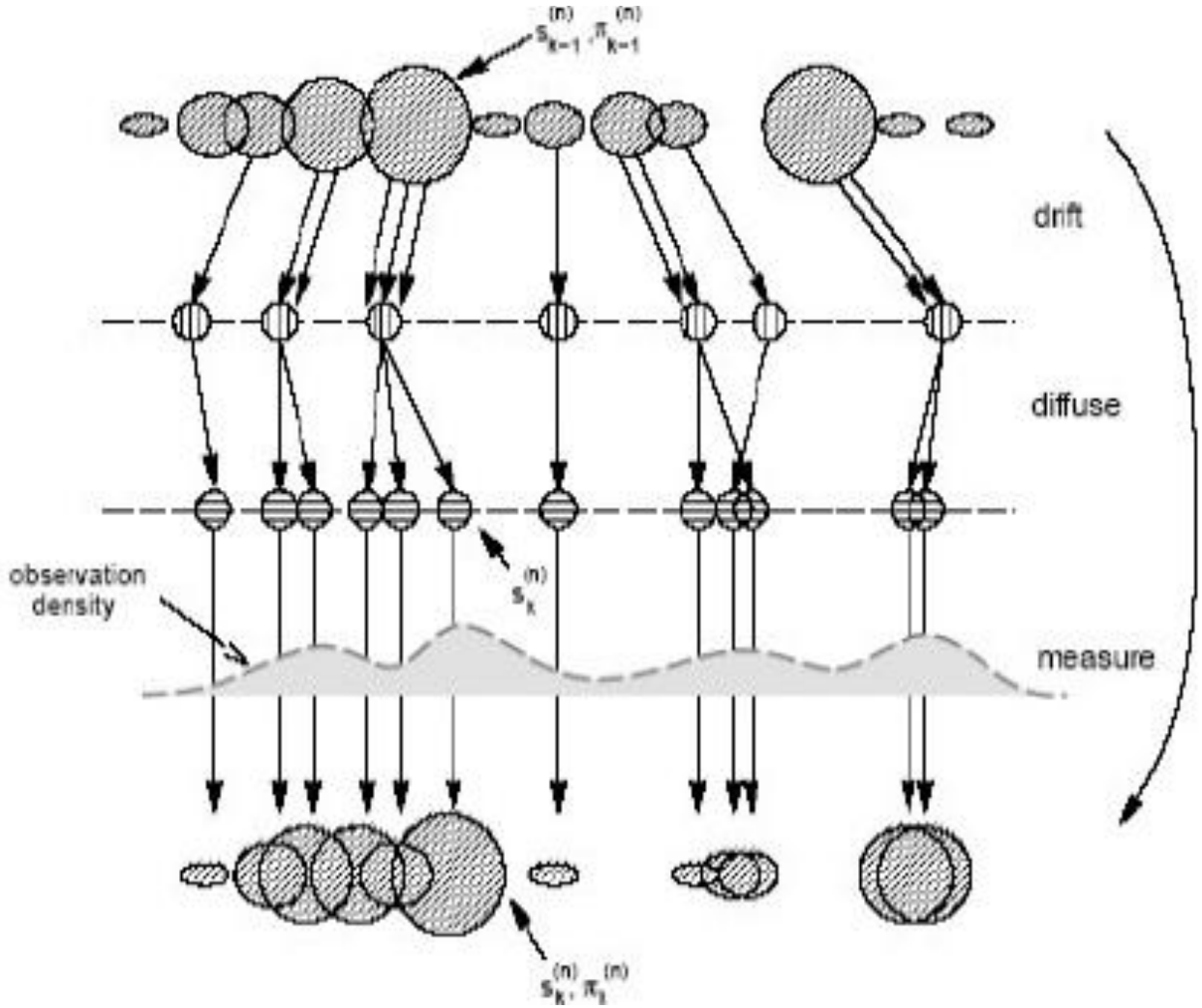
Particle Filters



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